CSCI 4302 Homework 4 Write Up

Marcus Winton

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1 Bayes Filter

Approach

When building a Bayes filter there are several needed pieces

$$x - \text{Robot State vector}$$
$$u - \text{Motion model}$$
$$z - \text{Measurement vector}$$

Using these pieces running the Bayes filter is fairly straight forward and does not require complex derivations of matrices, unlike the EKF. Instead the there is simply and iterative process of predicting what the model should do and then updating this prediction based on sensor data, using Bayes rule to calculate the most probable state given current readings and predictions.

Challenges

Overall implementing the Bayes filter was fairly straight forward. I ran into a small bug with implementing the update state, where I was multiplying the likelihood over the headings instead of averaging them, which obliviously lead to erroneous results.

Results

Given the above implementation the result was as follows

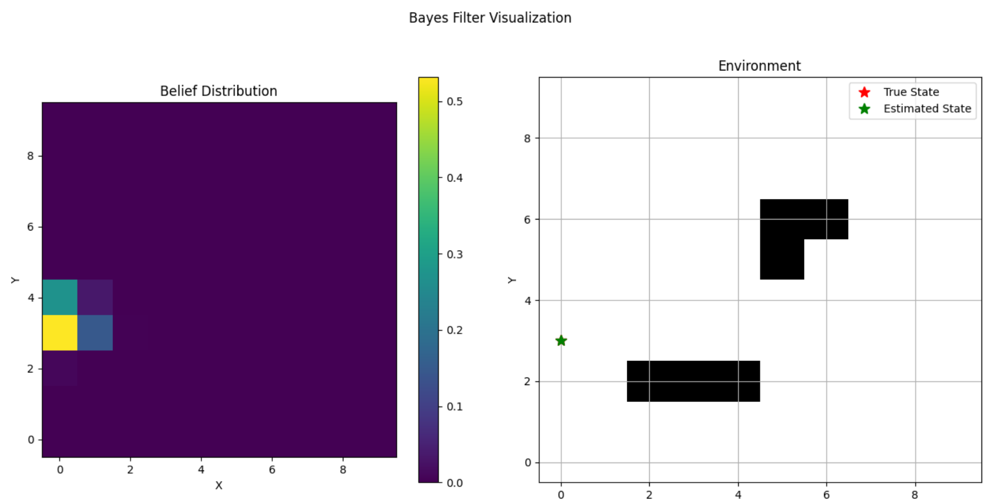

Figure 1: Bayes filter dynamics visualizer

2 Extended Kalman Filter

Approach

When tackling the EKF, similar to the Bayes filter, I started by familiarizing myself with the code base, and by trying to get what was needed for the EKF. The Extended Kalman Filters requires several matrices in order to work properly.

$$\begin{aligned}
\mathbf{x} &- \text{The state vector} \\
f(\mathbf{x}, \mathbf{u}) &- \text{Process model} \\
h(\mathbf{x}) &- \text{Measurement model} \\
F &- \text{Process model Jacobian} \\
H &- \text{Measurement model Jacobian} \\
Q &- \text{Process noise covariance} \\
R &- \text{Measurement noise covariance} \\
P &- \text{State noise covariance} \\
K &- \text{Kalman gain matrix}
\end{aligned}$$

In order to actually implement this however we need to derivative the two Jacobian matrices based on the processes and measurement models. Starting with the process model

$$F = \frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial x_{next}}{\partial x} & \frac{\partial x_{next}}{\partial y} & \frac{\partial x_{next}}{\partial \theta} \\ \frac{\partial y_{next}}{\partial x} & \frac{\partial y_{next}}{\partial y} & \frac{\partial y_{next}}{\partial \theta} \\ \frac{\partial \theta_{next}}{\partial x} & \frac{\partial \theta_{next}}{\partial y} & \frac{\partial \theta_{next}}{\partial \theta} \end{bmatrix}$$

Let

$$u_v = \text{linear velocity}, \qquad u_\omega = \text{angular velocity}, \qquad s(x, y) = \text{velocity scaling}, \qquad s_x = \frac{\partial}{\partial x} s(x, y), \qquad s_y = \frac{\partial}{\partial y} s(x, y)$$

$$
\begin{aligned}
\frac{\partial x_{next}}{\partial x} &= \frac{\partial}{\partial x}\left[x + u_v \cdot s(x,y) \cdot \cos(\theta) \cdot dt\right] \\
&= 1 + u_v \cdot s_x \cdot \cos(\theta) \cdot dt \\
\frac{\partial x_{next}}{\partial y} &= \frac{\partial}{\partial y}\left[x + u_v \cdot s(x,y) \cdot \cos(\theta) \cdot dt\right] \\
&= u_v \cdot s_y \cdot \cos(\theta) \cdot dt \\
\frac{\partial x_{next}}{\partial \theta} &= \frac{\partial}{\partial \theta}\left[x + u_v \cdot s(x,y) \cdot \cos(\theta) \cdot dt\right] \\
&= -v_{eff} \cdot \sin(\theta) \cdot dt \\
\frac{\partial y_{next}}{\partial x} &= \frac{\partial}{\partial y}\left[y + u_v \cdot s(x,y) \cdot \sin(\theta) \cdot dt\right] \\
&= u_v \cdot s_x \cdot \sin(\theta) \cdot dt \\
\frac{\partial y_{next}}{\partial y} &= \frac{\partial}{\partial y}\left[y + u_v \cdot s(x,y) \cdot \sin(\theta) \cdot dt\right] \\
&= 1 + u_v \cdot s_y \cdot \sin(\theta) \cdot dt \\
\frac{\partial y_{next}}{\partial \theta} &= \frac{\partial}{\partial \theta}\left[y + u_v \cdot s(x,y) \cdot \cos(\theta) \cdot dt\right] \\
&= -v_{eff} \cdot \cos(\theta) \cdot dt \\
\frac{\partial \theta_{next}}{\partial q} &= \frac{\partial}{\partial q}\left[\theta + \omega_{eff}(x,y) \cdot dt\right], \quad q \in \{x, y\} \\
&= \frac{\partial \omega_{eff}}{\partial q} dt \\
\omega_{eff} &= \frac{u_\omega}{\text{speed factor}} \\
\frac{\partial \omega_{eff}}{\partial q} &= u_\omega \left(-\frac{1}{\text{speed factor}^2} \cdot \frac{\partial \text{speed factor}}{\partial q}\right) \\
\text{speed factor} &= \frac{|v_{eff}|}{v_{max}} \\
\frac{\partial \text{speed factor}}{\partial q} &= \frac{1}{v_{max}} \sigma u_v s_q, \text{ where } \sigma = \text{sign}(v_{eff}) \\
\frac{\partial \theta_{next}}{\partial q} &= -u_\omega \cdot \sigma \frac{u_v}{v_{max}} \cdot \frac{s_q}{\text{speed factor}^2} \cdot dt \\
\frac{\partial \theta_{next}}{\partial \theta} &= \frac{\partial}{\partial \theta}\left[\theta + \omega_{eff}(x,y) \cdot dt\right] \\
&= 1
\end{aligned}
$$

Thus

$$
J = \begin{bmatrix}
1 + u_v \cdot s_x \cdot \cos(\theta) \cdot dt & u_v \cdot s_y \cdot \cos(\theta) \cdot dt & -v_{eff} \cdot \sin(\theta) \cdot dt \\
u_v \cdot s_x \cdot \sin(\theta) \cdot dt & 1 + u_v \cdot s_y \cdot \sin(\theta) \cdot dt & -v_{eff} \cdot \cos(\theta) \cdot dt \\
-u_\omega \cdot \sigma \frac{u_v}{v_{max}} \cdot \frac{s_x}{\text{speed factor}^2} \cdot dt & -u_\omega \cdot \sigma \frac{u_v}{v_{max}} \cdot \frac{s_y}{\text{speed factor}^2} \cdot dt & 1
\end{bmatrix}
$$

Moving on to the measurement Jacobian

$$
H = \frac{\partial h}{\partial x} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} & \frac{\partial r}{\partial \theta} \end{bmatrix}
$$

$$
\begin{aligned}
\frac{\partial r}{\partial x} &= \frac{\partial}{\partial x}\left[\sqrt{dx^2+dy^2}\right] \\
&= \frac{1}{2\sqrt{dx^2+dy^2}}\frac{\partial}{\partial x}\left(dx^2+dy^2\right) \\
&= \frac{2dx}{2\sqrt{dx^2+dy^2}} \\
&= \frac{dx}{r} \\
\frac{\partial r}{\partial y} &= \frac{\partial}{\partial y}\left[\sqrt{dx^2+dy^2}\right] \\
&= \frac{1}{2\sqrt{dx^2+dy^2}} \\
&= \frac{\partial}{\partial y}\left(dx^2+dy^2\right) \\
&= \frac{2dy}{2\sqrt{dx^2+dy^2}} \\
&= \frac{dy}{r} \\
\frac{\partial r}{\partial \theta} &= \frac{\partial}{\partial \theta}\left[\sqrt{dx^2+dy^2}\right] \\
&= 0
\end{aligned}
$$

Thus

$$H = \begin{bmatrix} \frac{dx}{r} & \frac{dy}{r} & 0 \end{bmatrix}$$

Once these matrices were calculated it was actually pretty simple to get working.

Challenges

One of the main challenges was initially I had made an error in calculating the process model Jacobian, J. The error in my calculation led to massive heading error, over 50°. However, the instructions in this assignment similar to the Bayes filter was straight forward and relevantly painless to implement.

Results

With all of the above information running the Extended Kalman Filter provides the following

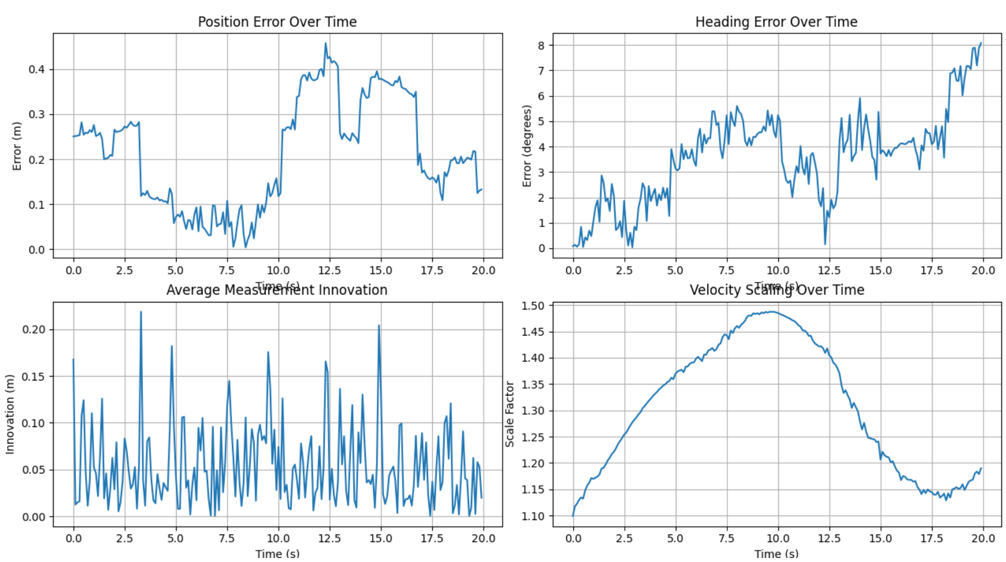

Figure 2: EKF error over time

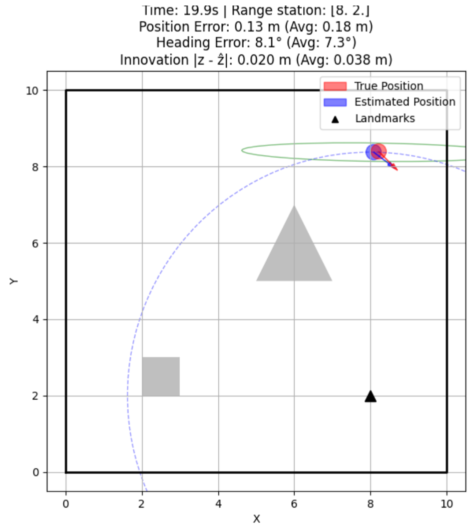

Figure 3: Final position estimate from EKF